

Week - 02 - Arithmetic, Strings, Lists\lecture_3_code.py

```
1  # def functiona_name (inputs)
2  # def find_quadratic_roots(paramters):
3  #     # write your quadratic formula solution here
4
5  import math      # math module
6  # we have written the value of pi in math module
7
8  def calculate_square_area(length):
9      '''
10     ( This comment is called a docstring )
11     This function calculates area of a square.
12     Input: it takes length of a side
13     Output: It returns the area
14     '''
15     area = pow(length, 2)
16
17     return area
18
19 def calculate_rectangle_area(width, height):
20     '''
21     This function calculates area of a rectangle
22     '''
23     return width * height
24
25 def calculate_triangle_area(base, height):
26     return base * height / 2
27
28 def calculate_circle_area(radius):
29     return 2 * math.pi * radius
30
31 def UI():
32     while True:
33         print("Dear user: Please choose which shape area you want to calculate: ")
34         print(" Shape Area Calculator\n" \
35             "1. Calculate square area\n" \
36             "2. Calculate rectangle area\n" \
37             "3. Calculate triangle area\n" \
38             "4. Calculate circle area\n" \
39             "5. Calculate cylinder surface area\n" \
40             "6. Compare two shapes\n" \
41             "7. Quit")
42         user_choice = int(input("Enter your choice: "))
43         if user_choice == 1:
44             side = float(input("Enter side measurement: "))
45             area = calculate_square_area(side)
46             print(f'Area of square is {area}')
47
48         elif user_choice == 2:
```

```

49         pass
50
51     elif user_choice == 7:
52         break
53
54 def UI2():
55     while True:
56         print(" Shape Area Calculator\n" \
57             "1. Calculate square area\n" \
58             "2. Calculate rectangle area\n" \
59             "3. Calculate triangle area\n" \
60             "4. Calculate circle area\n" \
61             "5. Calculate cylinder surface area\n" \
62             "6. Compare two shapes\n" \
63             "7. Quit")
64
65         area = 0.0
66
67         choice = int(input("Please enter your choice: "))
68
69         if choice == 7:
70             break
71         elif choice == 6:
72             radius = float(input("Enter radius: "))
73             circle_area = calculate_circle_area(radius)
74
75             base = float (input("Enter base: "))
76             height = float (input("Enter height: "))
77             triangle_area = calculate_triangle_area(base, height)
78
79             if circle_area == triangle_area:
80                 print(f"Both shapes have same area {circle_area} and {triangle_area}")
81             else:
82                 print("Both shapes have different area")
83
84         elif choice == 5:
85             pass
86         elif choice == 4:
87             radius = float(input("Enter radius: "))
88             area = calculate_circle_area(radius)
89         elif choice == 3:
90             base = float (input("Enter base: "))
91             height = float (input("Enter height: "))
92             area = calculate_triangle_area(base, height)
93         elif choice == 2:
94             width = float (input("Enter width: "))
95             height = float (input("Enter height: "))
96             area = calculate_rectangle_area(width, height)
97         elif choice == 1:
98             side = float (input ("Enter side length: "))

```

```
99         area = calculate_square_area
100     else:
101         print(f"You have entered an incorrect choice {choice}: Please try again")
102
103     print(f'The area of your shape is {area:.3f}')
104
105 UI()
106
107 # area = calculate_rectangle_area(width, height)
108 # print(area)
109
110
111
112
113 # length = float(input("Enter length: "))
114
115 # area = calculate_square_area(length)
116 # print("The area is:", area, " cm^2")
117 # # area = round(area, 3)
118 # print(area)
119 # print(f"The rounded area is {area:.2f} cm^2" )
120 # print(area)
```